Sahaj Bhadja

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Summary

Software engineer with 2+ years of experience building **cloud-backed full-stack systems**, **CI/CD workflows**, and **production data services** across AI and robotics-adjacent platforms using **AWS, Kubernetes, Docker, REST, SQL**, and automated testing.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Web Technologies, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Architected a **multi-tenant AI platform** on **Azure SQL** and Blob Storage with row-level security for **500+ users**, reducing dashboard latency from **1.2s to 350ms** through query and indexing optimization.
- Built **FastAPI** integration and release workflows across **PostgreSQL**, SQL Server, and cloud storage, processing **5K+ monthly jobs** with validation checks, incident debugging, and **99%+ production reliability**.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Developed and tested a **Flask + React** grading platform handling **10K+ submissions**, integrating secure file ingestion, rubric workflows, and multimodal model execution for **150+ instructors**.
- Designed concurrent backend pipelines on **AWS RDS** with multithreading, **REST APIs**, and **GitHub Actions CI/CD**, sustaining **10K+ grading requests/day** while reducing turnaround time by **60%**.

USC AD Labs **Los Angeles, California**
ML Research Assistant *January 2025 – May 2025*
- Built simulation and model-training pipelines for autonomous-driving behavioral planning, processing **100K+ trajectory logs** and improving long-horizon planning quality by **18%**.
- Architected an HD-map-aware spatial-temporal encoder in **PyTorch** that improved closed-loop planning stability by **15%**, strengthening robotics and large-scale evaluation workflow experience.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform **microservice** software with **Jenkins CI/CD**, PyTest, and Mockito, automating builds, deployments, and **20+ unit/integration suites** across Windows, Linux, and macOS.
- Optimized API and ETL workflows with **AWS Glue**, **S3**, and OOP patterns, improving response times by **30%**, boosting pipeline efficiency by **68%**, and reducing technical debt by **20%**.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a full-stack **MERN** resource-management platform with **REST APIs**, WebSockets, and Dockerized microservices, supporting **8+ workflows** and **2M+ synchronized records** across locations.
- Automated **Jenkins**-based testing and deployments for **Azure SQL**-backed services, improving release reliability while reducing resource wastage by **42%** and increasing lead conversion by **15%**.

Skills

Languages: Python, Java, C++, JavaScript/TypeScript, SQL, Golang
Backend & APIs: FastAPI, Flask, Node.js, Express.js, REST APIs, Microservices, Message Brokering
Frontend: React.js, Next.js, TailwindCSS
Databases: PostgreSQL, MySQL, MongoDB, SQL Server, Azure SQL
Cloud & Infrastructure: AWS (EC2, S3, RDS, Lambda), Microsoft Azure, Docker, Kubernetes, GitHub Actions, Jenkins
AI/ML: PyTorch, Transformers, NLP, LLM Systems, Feature Engineering
Engineering Practices: CI/CD, Unit & Integration Testing, Distributed Systems, ETL Pipelines, Performance Optimization, Technical Documentation, Agile

Projects

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB, Plaid*
- Built a scalable fintech platform supporting **50K+ users** and **1M+ transactions** with **Kubernetes**, real-time updates, and secure tokenized APIs for cloud-native deployment and operational reliability.

Chat2SQL Live | *Java, TypeScript, SQL, Docker*
- Built an AI-powered SQL workspace supporting **10+ databases**, combining Java + TypeScript services, Dockerized deployment, schema exploration, and AI-assisted query generation from a unified interface.

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL*
- Built a semantic document platform over **50K+ documents** with sub-second retrieval, secure collaboration, and cloud indexing pipelines that cut document review time by **60%**.